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Infections of the Urinary Tract

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Keywords

antibiotic prophylaxis for urologic procedures; asymptomatic bacteriuria; cystitis; urinary microbiome; urinary tract infections

Questions

1. Acute pyelonephritis is the most likely diagnosis in a patient with:
    a. chills, fever, and flank pain.
    b. bacteria and pyuria.
    c. focal scar in renal cortex.
    d. delayed renal function.
    e. vesicoureteral reflux.
2. Bacteriuria without pyuria is indicative of:
    a. infection.
    b. colonization.
    c. tuberculosis.
    d. contamination.
    e. stones.
3. Nosocomial urinary tract infections (UTIs):
    a. occur in patients who are hospitalized or institutionalized.
    b. are caused by common bowel bacteria.
    c. can be suppressed by low-dose antimicrobial therapy.
    d. are due to reinfection.

e. are due to bacterial persistence.

4. Most recurrent infections in female patients are:
    a. complicated.
    b. reinfections.
    c. due to bacterial resistance.
    d. due to hereditary susceptibility factors.
    e. composed of multiple organisms.
5. Rates of reinfection (i.e., time to recurrence) are influenced by:
    a. bladder dysfunction.
    b. renal scarring.
    c. vesicoureteral reflux.
    d. antimicrobial treatment.
    e. age.
6. The long-term effect of uncomplicated recurrent UTIs is:
    a. renal scarring.
    b. hypertension.
    c. azotemia.
    d. ureteral vesical reflux.
    e. minimal.
7. The ascending route of infection is least enhanced by:
    a. catheterization.
    b. spermicidal agents.
    c. indwelling catheter.
    d. fecal soilage of perineum.
    e. frequent voiding.
8. Approximately 10% of symptomatic lower UTIs in young, sexually active female patients are caused by:
    a. *Escherichia coli (E. coli)*.
    b. *Staphylococcus saprophyticus.*
    c. *Pseudomonas.*
    d. *Proteus mirabilis.*
    e. *Staphylococcus epidermidis.*
9. The virulence factor that is most important for adherence is:
    a. hemolysin.
    b. K antigen.
    c. pili.

d. colicin production.
e. O serogroup.

10. Phase variation of bacterial pili:
    a. occurs only in vitro.
    b. affects bacterial virulence.
    c. is characteristic of pyelonephritic *E. coli.*
    d. is irreversible.
    e. refers to change in pilus length.
11. The finding that first suggested a biologic difference in women susceptible to UTIs is:
    a. increased adherence of bacteria to vaginal cells.
    b. decreased estrogen concentration in vaginal cells.
    c. elevated vaginal pH.
    d. nonsecretor status.
    e. postmenopausal status.
12. The primary bladder defense is:
    a. low urine pH.
    b. low urine osmolarity.
    c. voiding.
    d. Tamm-Horsfall protein (uromucoid).
    e. vaginal mucus.
13. The most significant sequela of renal papillary necrosis is renal:
    a. failure.
    b. abscess.
    c. obstruction.
    d. stone.
    e. cancer.
14. Severity and morbidity of bacteriuria is most morbid in patients with:
    a. spinal cord injuries.
    b. pregnancy.
    c. reflux.
    d. diabetes mellitus.
    e. human immunodeficiency virus (HIV) infection.
15. The validity of a midstream urine specimen should be questioned if microscopy reveals:

a. squamous epithelial cells.
b. red blood cells.
c. bacteria.
d. white blood cells.
e. casts.

16. Urinary tract imaging is NOT usually indicated for recurrent UTIs in:
    a. women.
    b. girls.
    c. men.
    d. boys.
    e. spinal cord–injured patients.
17. The most sensitive imaging modality for diagnosing renal abscess is:
    a. ultrasonography.
    b. indium scanning.
    c. gallium scanning.
    d. excretory urography.
    e. CT.
18. Treatment of UTIs depends most on an antimicrobial agent's:
    a. serum half-life.
    b. serum level.
    c. urine level.
    d. duration of therapy.
    e. frequency of therapy.
19. An ideal class of drugs for the treatment of uncomplicated symptomatic UTIs in women is:
    a. aminopenicillins.
    b. aminoglycosides.
    c. fluoroquinolones.
    d. cephalosporins.
    e. nitrofurantoins.
20. The host factor least likely to be associated with an increased risk of infection is:
    a. advanced age.
    b. a history of previous infection in the site/organ of interest.

c. residence in a chronic care facility.
d. indwelling orthopedic pins.
e. coexistent infection.

21. The optimal duration of antimicrobial therapy with Bactrim for symptomatic acute uncomplicated cystitis in women is:
    a. 1 day.
    b. 3 days.
    c. 7 days.
    d. 14 days.
    e. 21 days.
22. Treatment of asymptomatic bacteriuria is most indicated in patients who are:
    a. elderly.
    b. catheterized.
    c. pregnant.
    d. confused.
    e. incontinent.
23. Screening for bacteriuria is beneficial in:
    a. pregnant women.
    b. elderly patients.
    c. men.
    d. children.
    e. spinal cord–injured patients.
24. The most common cause of unresolved bacteriuria during antimicrobial therapy is:
    a. development of bacterial resistance.
    b. rapid reinfections.
    c. azotemia.
    d. staghorn calculi.
    e. initial bacterial resistance.
25. Nitrofurantoin is effective because of the concentration of the drug in the:
    a. urine.
    b. vaginal mucus.
    c. bowel.
    d. serum.

e. bladder.

26. The most common cause of acute pyelonephritis in young women is:
    a. vesicoureteral reflux.
    b. P-piliated bacteria.
    c. type 1 piliated bacteria.
    d. recurrent UTIs.
    e. bacterial endotoxin.
27. An optimal oral antibiotic agent for the treatment of acute uncomplicated pyelonephritis in a pregnant women is:
    a. TMP-SMX
    b. Cephalexin
    c. Amoxicillin
    d. Levofloxacin
    e. Macrobid
28. A patient with acute pyelonephritis, persistent fever, and flank pain for 24 hours warrants:
    a. observation.
    b. CT.
    c. change in antimicrobial therapy.
    d. ultrasonography.
    e. blood cultures.
29. Emphysematous pyelonephritis usually occurs in:
    a. children
    b. adults with a history of renal transplant
    c. women with a history of recurrent uncomplicated UTIs
    d. diabetic adults
    e. adults on clean intermittent catheterization
30. The primary treatment for a small perirenal abscess in a functioning kidney is:
    a. nephrectomy.
    b. partial nephrectomy.
    c. open surgical drainage.
    d. percutaneous drainage.
    e. retrograde ureteral drainage.
31. Most patients with chronic pyelonephritis present with:

a. hypertension.
b. renal failure.
c. chronic infection.
d. flank pain.
e. no symptoms.

32. The most common bacterial cause of xanthogranulomatous pyelonephritis is:
    a. *E. coli.*
    b. *Pseudomonas.*
    c. *Klebsiella.*
    d. *Proteus mirabilis.*
    e. *Staphylococcus.*
33. Michaelis-Gutmann bodies are associated with the following disease process:
    a. Xanthogranulomatous pyelonephritis
    b. Malacoplakia
    c. Renal echinococcosis
    d. Chronic pyelonephritis
    e. Acute focal bacterial nephritis
34. Treatment of renal echinococcosis involves which of the following:
    a. treatment with antibiotics and follow-up imaging to confirm regression of the hydatid cyst
    b. observation
    c. aspiration of cyst contents
    d. surgical removal of the hydatid cyst
    e. injection of the hydatid cyst with targeted antibiotics
35. The most reliable early clinical indicator of septicemia is:
    a. chills.
    b. fever.
    c. hyperventilation.
    d. lethargy.
    e. change in mental status.
36. Compared with non-pregnant women, pregnant women have a higher prevalence of:
    a. asymptomatic bacteriuria.
    b. acute cystitis.

c. acute pyelonephritis.
d. recurrent cystitis.
e. bacterial persistence.

37. Clinical pyelonephritis during pregnancy is most commonly linked to:
    a. maternal sepsis.
    b. maternal anemia.
    c. maternal hypertension.
    d. eclampsia.
    e. congenital malformations.
38. The drug thought to be safe in any phase of pregnancy is:
    a. a fluoroquinolone.
    b. nitrofurantoin.
    c. a sulfonamide.
    d. penicillin.
    e. tetracycline.
39. The majority of elderly patients with bacteriuria are:
    a. asymptomatic.
    b. febrile.
    c. incontinent.
    d. confused.
    e. dysuric.
40. In the absence of obstruction, treatment of asymptomatic bacteriuria in the elderly:
    a. is cost effective.
    b. prevents renal failure.
    c. reduces mortality.
    d. reduces morbidity.
    e. is unnecessary.
41. Prophylaxis for endocarditis should be administered in patients with:
    a. a history of childhood heart murmurs.
    b. heart valves inserted more than 5 years ago.
    c. calcified heart valves associated with a murmur.
    d. all synthetic heart valves.
    e. cadaveric heart valves.

f. none of the above.

42. The most common predisposing factor for hospital-acquired UTIs is:
    a. surgery.
    b. antimicrobial therapy.
    c. age.
    d. catheterization.
    e. diabetes mellitus.
43. The most effective measure for reducing catheter-associated UTI is:
    a. closed drainage.
    b. antimicrobial prophylaxis.
    c. catheter irrigation.
    d. intermittent catheterization.
    e. daily meatal care.
44. In spinal cord–injured patients the bladder drainage technique with the lowest complication rate is:
    a. clean intermittent catheterization (CIC).
    b. suprapubic drainage.
    c. indwelling catheter.
    d. condom catheter.
    e. suprapubic pressure.
45. All of the following conditions are predisposing factors to the development of Fournier gangrene EXCEPT:
    a. obesity
    b. paraphimosis
    c. diabetes mellitus
    d. perirectal infections
    e. urethral strictures
46. Which of the following is true of vaginal estrogen preparations and recurrent urinary tract infections in postmenopausal women?
    a. The use of vaginal estrogen has been associated with an increased risk of breast cancer.
    b. Vaginal estrogen use can confer an increased risk of thrombotic events in women.
    c. Vaginal estrogen preparations can help with vaginal pain symptoms that can be conflated for UTI but do not modulate

UTI risk independently.

d. Vaginal estrogen can prevent recurrent UTIs and these effects are immediate.

e. The effects of vaginal estrogen on recurrent UTI risk are related to modulation of local pH and vaginal microflora.

47. Which of the following is not a risk factor for UTI in a renal transplant recipient?
    a. Cadaveric graft
    b. Diabetes
    c. Prolonged hemodialysis prior to transplant
    d. Female gender
    e. Polycystic native kidneys
48. Prophylaxis for endocarditis should be administered in patients with:
    a. a history of childhood heart murmurs.
    b. heart valves inserted more than 5 years ago.
    c. calcified heart valves associated with a murmur.
    d. all synthetic heart valves.
    e. cadaveric heart valves.
    f. none of the above.
49. Antimicrobial prophylaxis is characterized as:
    a. administration of an antimicrobial agent within 4 to 6 hours of the procedure.
    b. administration of an antimicrobial agent for a period of time covering the first 48 hours after the procedure.
    c. administration of an antimicrobial agent within 30 minutes of the initiation of a procedure and for a period of time covering the first 48 hours after the procedure.
    d. administration of an antimicrobial agent within 60 to 120 minutes of the initiation of a procedure and for a period of time that covers the duration of the procedure.
    e. administration of an antimicrobial agent the night before the initiation of a procedure and for a period of time that covers the duration of the procedure.
50. Which of the following organisms is NOT associated with positive nitrites on urine analysis?

a. *Enterococcus*
b. *E. coli*
c. *Proteus mirabilis*
d. *Klebsiella pneumoniae*

51. All of the following are important factors to consider in a patient's ability to give a noncontaminated midstream urine sample except:
    a. Vaginal atrophy
    b. Poor manual dexterity
    c. History of human papilloma virus (HPV)
    d. Inability to weight beat
    e. Presence of a vaginal pessary
52. Which of the following is least relevant to history-taking when interviewing a patient with recurrent UTIs?
    a. Family history of postmenopausal UTIs
    b. Spermicide use
    c. Childhood voiding dysfunction
    d. History of urologic surgery
    e. Past urinary pathogens and antibiotics

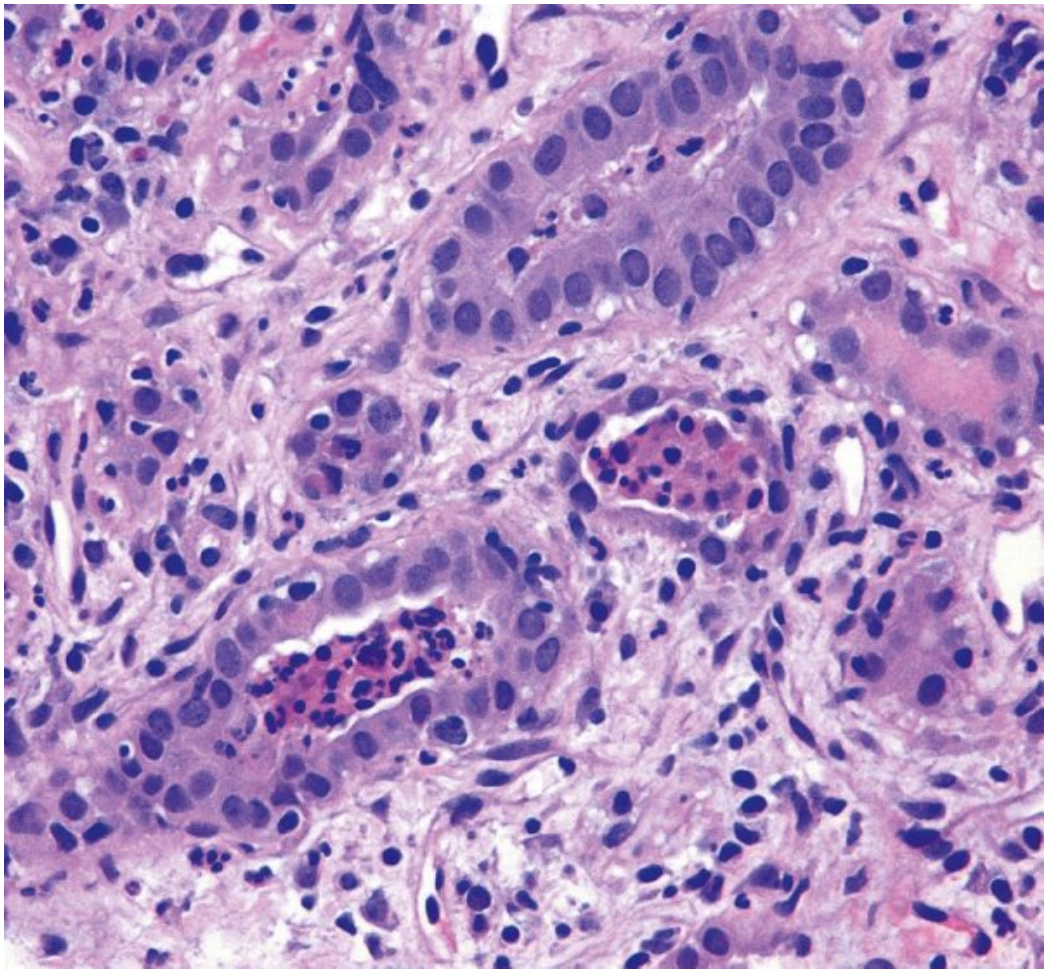


FIG. 55.1

Pathology

1. See Fig. 55.1. A 65-year-old woman has the acute onset of right flank pain, fever, and an enlarged kidney on imaging. Blood cultures and urine cultures are obtained and broad-spectrum antibiotics administered. The patient improves, but the kidney on imaging remains enlarged. A needle biopsy of the kidney is obtained. The pathology report is acute pyelonephritis with numerous neutrophils within the interstitium and the renal tubules. The biopsy:
    a. provides information as to the length of time antibiotics should be administered.

b. suggests that the antibiotics should be changed.
c. is unnecessary.
d. suggests the need for a percutaneous drain.
e. suggests that an abscess is likely to develop.

2. A 65-year-old man has fever and malaise. A CT scan reveals an 8-cm solid mass in his left kidney with marked thickening of the retroperitoneum around the kidney and pancreas. The kidney is poorly functioning and there is a 1-cm stone in the renal pelvis. A biopsy is done and reveals xanthogranulomatous pyelonephritis, which is depicted in Fig. 55.2. The next step in management is:
   a. extracorporeal shockwave lithotripsy.
   b. biopsy of the retroperitoneum.
   c. left nephrectomy.
   d. urine culture and treatment according to sensitivities.
   e. partial left nephrectomy
3. A 45-year-old woman is found to have a raised bladder lesion on cystoscopy. The biopsy shown in Fig. 55.3 reveals malakoplakia. The next step in management is:
   a. intravesical bacille Calmette-Guérin.
   b. fulguration of the lesions.
   c. intravesical mitomycin C.
   d. treat with a sulfonamide for several months.
   e. a 3-day course of ciprofloxacin.

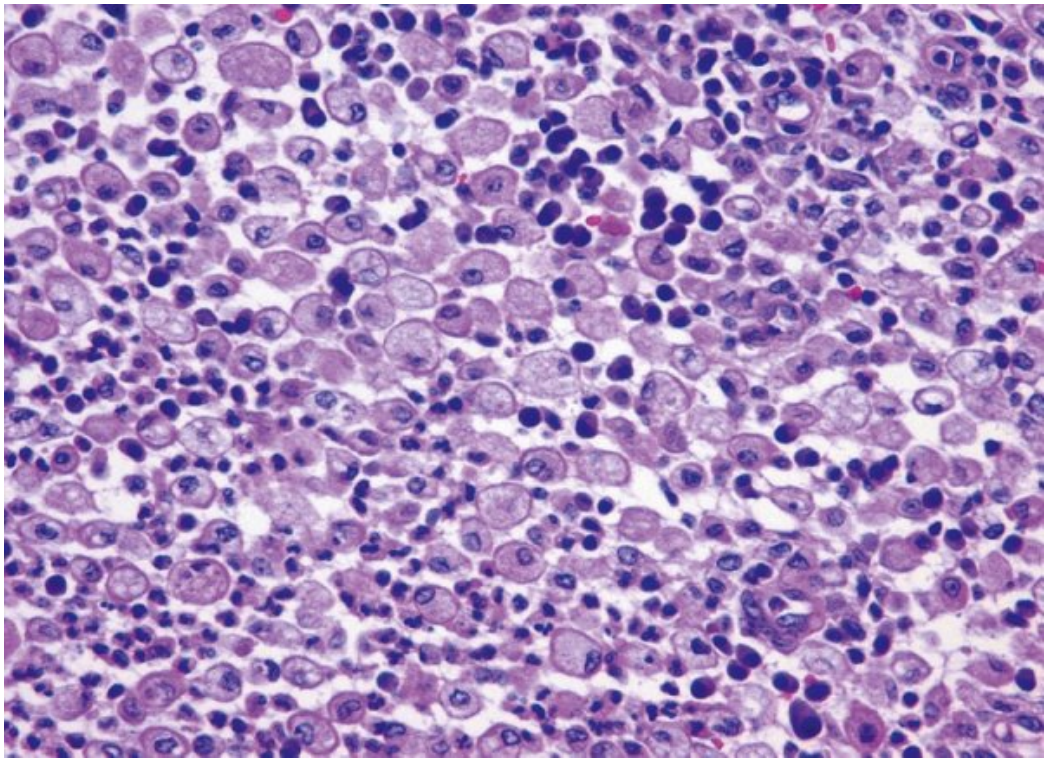


FIG. 55.2

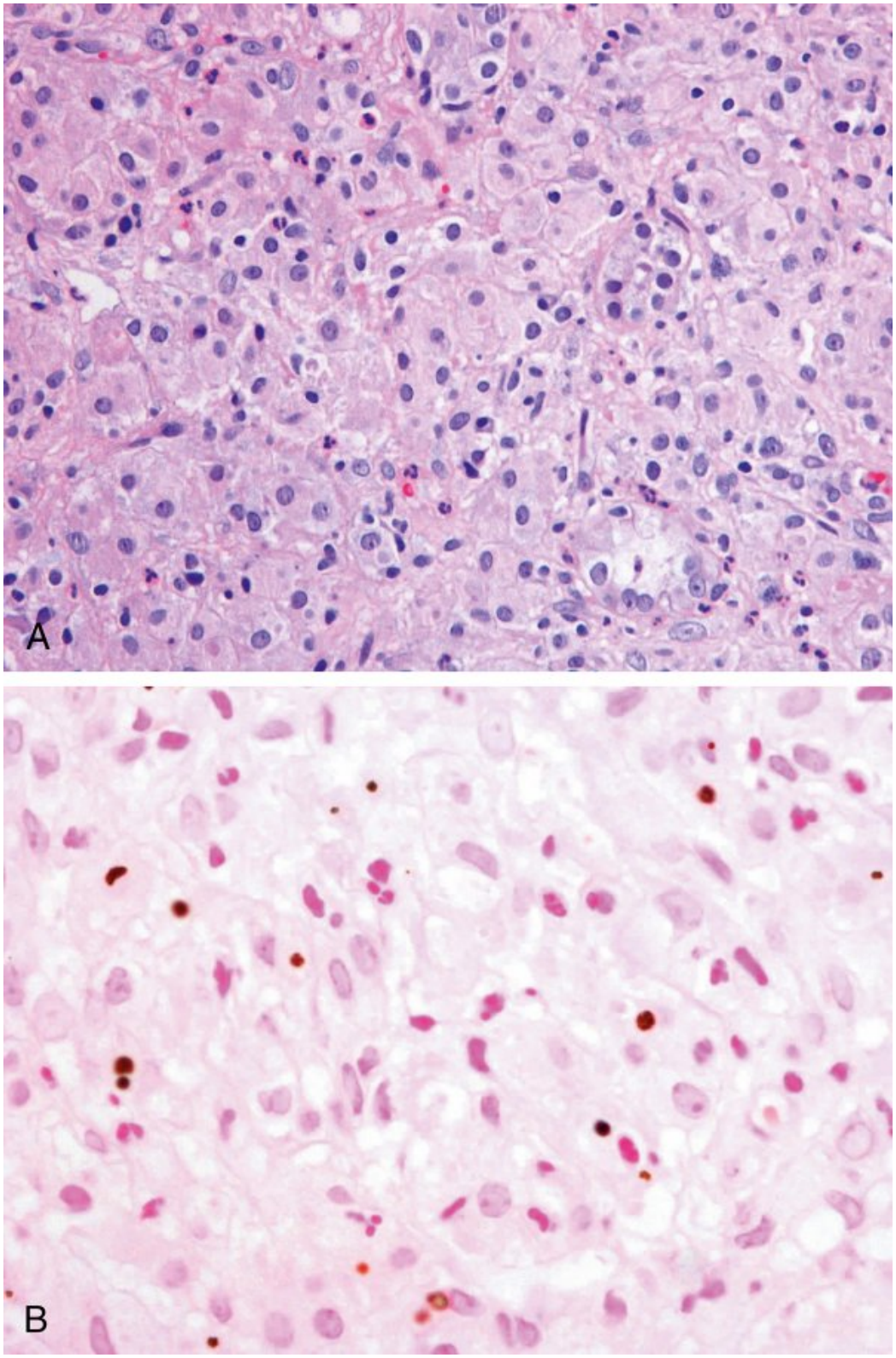


FIG. 55.3

Imaging

1. A 72-year-old man presents with right flank pain and fever. A contrast-enhanced CT scan is shown in Fig. 55.4. The most likely diagnosis is:
    a. acute right renal obstruction.
    b. delayed excretion in left kidney.
    c. cellulitis in right flank.
    d. right perinephric abscess.
    e. xanthogranulomatous pyelonephritis.

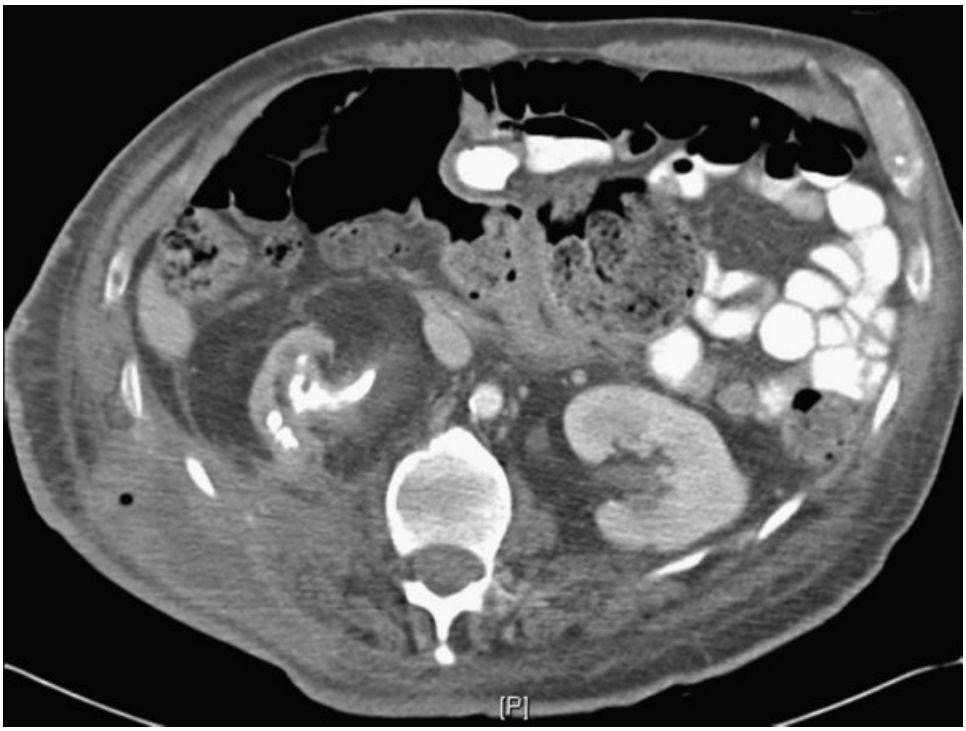


FIG. 55.4

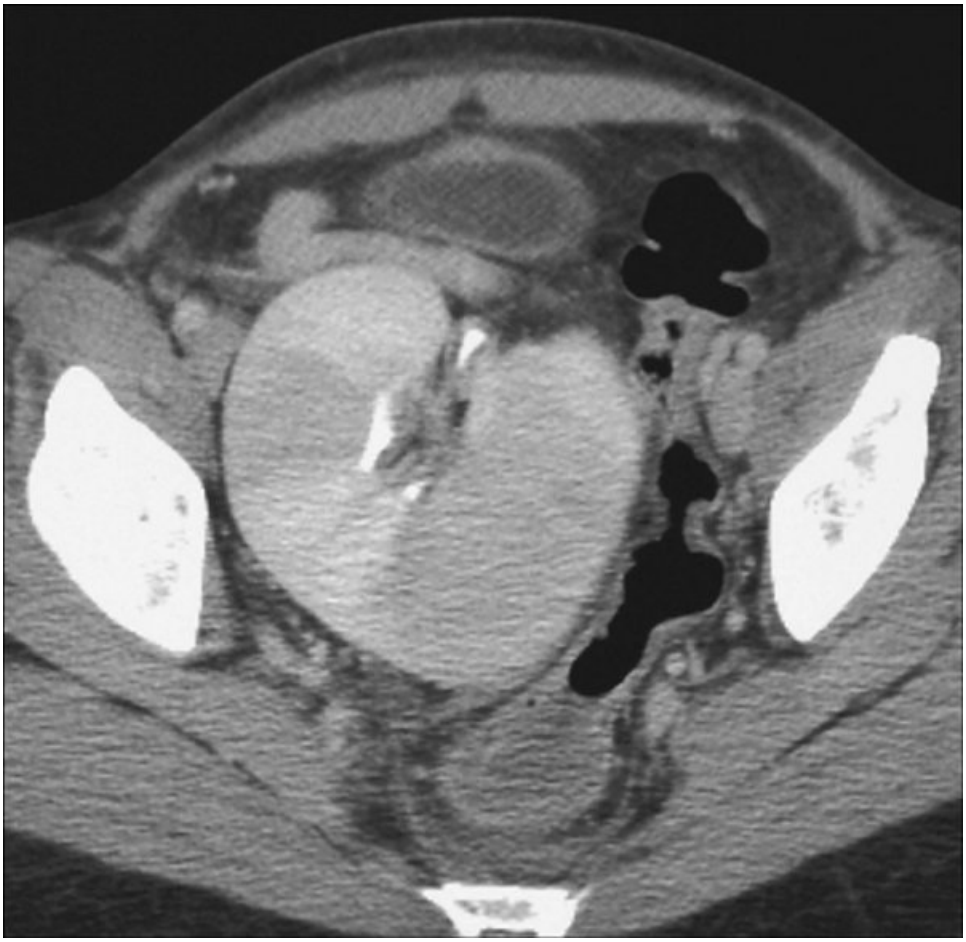


FIG. 55.5

2. A 40-year-old woman presents with pelvic pain and fever. A contrast-enhanced CT scan is shown in Fig. 55.5. The most likely diagnosis is:
    a. renal infarct.
    b. renal artery occlusion.
    c. chronic pyelonephritis.
    d. acute urinary obstruction.
    e. acute pyelonephritis.
3. A 22-year-old woman presents with shaking chills and fever. An enhanced CT image is shown in Fig. 55.6. The next step in management is:
    a. percutaneous drainage.
    b. nephrectomy.
    c. partial nephrectomy.

d. open surgical drainage.
e. cystoscopy and retrograde urography.

4. A 68-year-old diabetic woman presents to the emergency room with the chief complaint of abdominal pain. CT scan findings of her pelvis are shown in Fig. 55.7. The next step in her management is:



FIG. 55.6



FIG. 55.7

a. inpatient antibiotics, Foley catheter placement, and glucose management
b. collection of a urine sample and discharge home with empiric antibiotic coverage
c. emergent exploration and surgical debridement with placement of a suprapubic tube
d. obtain an ultrasound to assess the upper urinary tract
e. reassurance

Answers

1. **a. Chills, fever, and flank pain.** Acute pyelonephritis is a clinical syndrome of chills, fever, and flank pain that is accompanied by bacteriuria and pyuria, a combination that is reasonably specific for an acute bacterial infection of the kidney.
2. **b. Colonization.** Bacteriuria without pyuria is generally indicative of bacterial colonization without infection of the urinary tract.
3. **a. Occur in patients who are hospitalized or institutionalized.** Nosocomial or health care–associated UTIs occur in patients who are hospitalized or institutionalized and may be caused by *Pseudomonas* and other more antimicrobial-resistant strains.
4. **b. Reinfections.** Recurrent infections are hypothesized to be secondary to either bacterial persistence within the urinary tract or, more commonly, novel reinfection. Persistence, caused by the same bacterial strain, usually leads to recurrent infections in a short time frame, whereas reinfections generally occur over a more remote period. Reinfection is likely secondary to ascent of uropathogens from fecal flora into the urinary tract or from reemergence of bacteria from uroepithelial intracellular colonies.
5. **d. Antimicrobial treatment.** Whether a patient receives no treatment or short-term, long-term, or prophylactic antimicrobial treatment, the risk of recurrent bacteriuria remains the same; **antimicrobial treatment appears to alter only the time until recurrence.**
6. **e. Minimal.** The long-term effects of uncomplicated recurrent UTIs are not completely known, but so far, **no association between recurrent infections and renal scarring, hypertension, or progressive renal azotemia** has been established.
7. **e. Frequent voiding.** This route is further enhanced in individuals with significant soilage of the perineum with feces, women using spermicidal agents, and patients with intermittent or indwelling catheters.
8. **b. *Staphylococcus saprophyticus.*** *S. saprophyticus* is recognized as causing frequent symptomatic UTIs in young, sexually active females, whereas it rarely causes infection in males and elderly individuals.

9. **c. Pili.** Studies have demonstrated that interactions between FimH and receptors expressed on the luminal surface of the bladder epithelium are critical to the ability of many uropathogenic *E. coli* strains to colonize the bladder and cause disease.
10. **b. Affects bacterial virulence.** This process is called *phase variation* and has obvious biologic and clinical implications. For example, the presence of type 1 pili may be advantageous to the bacteria for adhering to and colonizing the bladder mucosa but disadvantageous because the pili enhance phagocytosis and killing by neutrophils.
11. **a. Increased adherence of bacteria to vaginal cells.** These studies established increased adherence of pathogenic bacteria to vaginal epithelial cells as the first demonstrable biologic difference that could be shown in women susceptible to UTI.
12. **c. Voiding.** Bacteria presumably make their way into the bladder fairly often. Whether small inocula of bacteria persist, multiply, and infect the host depends in part on the ability of the bladder to empty.
13. **c. Obstruction.** A patient who suffers from an acute ureteral obstruction caused by a sloughed papilla and who has a concomitant UTI should have the condition treated as a urologic emergency.
14. **a. Spinal cord injuries.** Of all patients with bacteriuria, no group compares in severity and morbidity with those who have spinal cord injury.
15. **a. Squamous epithelial cells.** The validation of the midstream urine specimen can be questioned if numerous squamous epithelial cells (indicative of preputial, vaginal, or urethral contaminants) are present.
16. **a. Women. Imaging and cystoscopic evaluation are not warranted in all women with recurrent UTIs.** Indeed, the yield of imaging in women without suspected complicated UTI is low and is not recommended by the American College of Radiology, the Canadian Urological Association Guidelines, or the European Association of Urology Guidelines. However, **in women with risk**

factors for a complicated UTI the evaluation should include imaging and cystoscopy.

17. **e. CT.** CT and magnetic resonance imaging are more sensitive than excretory urography or ultrasonography in the diagnosis of acute focal bacterial nephritis, renal and perirenal abscesses, and radiolucent calculi.
18. **c. Urine level.** Efficacy of the antimicrobial therapy is critically dependent on the antimicrobial levels in the urine and the length of time that this level remains above the minimum inhibitory concentration of the infecting organism. Thus resolution of infection is closely associated with the susceptibility of the bacteria to the concentration of the antimicrobial agent achieved in the urine.
19. **e. Nitrofurantoin.** According to the Infectious Diseases Society of America 2010 update, nitrofurantoin 100 mg twice daily for 5 days or Bactrim DS twice daily for 3 days should be preferential regimens for the treatment of uncomplicated UTIs in women. Sensitivity to these agents should be confirmed on urine culture, especially if the patient does not report a resolution of symptoms at the end of their course.
20. **d. Indwelling orthopedic pins.** Bacterial seeding of implanted orthopedic hardware is a rare but morbid event. A joint commission of the American Urological Association, the American Academy of Orthopaedic Surgeons, and infectious disease specialists convened in 2003 and released an advisory statement on antibiotic prophylaxis for urologic patients with total joint replacement. In general, **antimicrobial prophylaxis for urologic patients with total joint replacements, pins, plates, or screws is not indicated. Prophylaxis is advised for individuals at higher risk of seeding a prosthetic joint and include those with recently inserted implants (within 2 years).**
21. **b. 3 days.** According to IDSA guidelines, a 3-day therapy is the preferred regimen for uncomplicated cystitis in women.
22. **c. Pregnant.** In populations other than those for whom treatment has been documented to be beneficial (e.g., pregnant women and patients undergoing urologic interventions), screening for or

treatment of asymptomatic bacteriuria is not appropriate and should be strongly discouraged.

23. **a. Pregnant women.** In populations other than those for whom treatment has been documented to be beneficial (e.g., pregnant women and patients undergoing urologic interventions), screening for or treatment of asymptomatic bacteriuria is not appropriate and should be discouraged.
24. **e. Initial bacterial resistance.** Most commonly, the bacteria are resistant to the antimicrobial agent selected to treat the infection.
25. **a. Urine. Nitrofurantoin, which does not alter the bowel flora, is present for brief periods at high concentrations in the urine** and leads to repeated elimination of bacteria from the urine, presumably by interfering with bacterial initiation of infection.
26. **b. P-piliated bacteria.** If vesicourethral reflux is absent, a patient bearing the P blood group phenotype may have special susceptibility to recurrent pyelonephritis caused by *E. coli* that have P pili and bind to the P blood group antigen receptors.
27. **b. Cephalexin.** Macrobid is concentrated in the urine and cannot treat blood-borne infections. Amoxicillin has been used to treat cystitis in pregnancy but does not have broad enough gram-negative coverage for the treatment of pyelonephritis. Levofloxacin is contraindicated in pregnancy due to possible damage to fetal cartilage, and trimethoprim should be avoided in pregnancy because it may cause fetal megaloblastic anemia, and, in the first trimester, neural tube and cardiovascular defects by inhibiting folic acid metabolism.
28. **a. Observation.** Even though the urine usually becomes sterile within a few hours of starting antimicrobial therapy, patients with acute uncomplicated pyelonephritis may continue to have fever, chills, and flank pain for several more days after initiation of successful antimicrobial therapy. They should be observed.
29. **d. Diabetic adults.** Emphysematous pyelonephritis predominantly affects female diabetics and can occur in insulin-dependent and non-insulin-dependent patients in the absence of ureteral obstruction. Nondiabetic patients can also develop this form of

pyelonephritis but often have ureteric obstruction and do not seem to develop extensive disease

30. **d. Percutaneous drainage.** Although surgical drainage, or nephrectomy if the kidney is nonfunctioning or severely infected, is the classic treatment for perinephric abscesses, renal ultrasonography and CT make percutaneous aspiration and drainage of small perirenal collections possible.
31. **e. No symptoms.** There are no symptoms of chronic pyelonephritis until it produces renal insufficiency, and then the symptoms are similar to those of any other form of chronic renal failure.
32. **d. *Proteus mirabilis.*** Although review of the literature shows ***Proteus*** **to be the most common organism involved with xanthogranulomatous pyelonephritis, *E. coli* is also common.**
33. **b. Malacoplakia.** Malacoplakia, from the Greek word meaning "soft plaque," is an unusual inflammatory disease that was originally described to affect the bladder. It is an inflammatory lesion described originally by Michaelis and Gutmann 1902. It was characterized by von Hansemann 1903 as soft, yellow-brown plaques with granulomatous lesions in which the histiocytes contain distinct basophilic lysosomal inclusion bodies or Michaelis-Gutmann bodies. Although its exact pathogenesis is unknown, malacoplakia probably results from abnormal macrophage function in response to a bacterial infection, which is most often E. coli.
34. **d. Surgical removal of the hydatid cyst.** Surgery remains the mainstay of treatment of renal echinococcosis. The cyst should be removed without rupture to reduce the chance of seeding, antigen reaction, and recurrence. If the cyst ruptures or cannot be removed and marsupialization is required, the contents of the cyst initially should be aspirated and filled with a scolicidal agent.
35. **c. Hyperventilation.** Even before temperature extremes and the onset of chills, bacteremic patients often begin to hyperventilate. Thus the earliest metabolic change in septicemia is a resultant respiratory alkalosis.
36. **c. Acute pyelonephritis.** Pyelonephritis develops in 1% to 4% of all pregnant women and in 20% to 40% of pregnant women with

untreated bacteriuria.

37. **a. Maternal sepsis.** Pregnant women with asymptomatic bacteriuria are at higher risk for developing a symptomatic UTI that results in adverse fetal sequelae, complications associated with bacteriuria during pregnancy, and pyelonephritis and its possible sequelae, such as sepsis in the mother. Therefore all women with asymptomatic bacteriuria should be treated.
38. **d. Penicillin.** The aminopenicillins and cephalosporins are considered safe and generally effective throughout pregnancy. In patients with penicillin allergy, nitrofurantoin is a reasonable alternative.
39. **a. Asymptomatic.** Most elderly patients with bacteriuria are asymptomatic.
40. **e. Is unnecessary. Prospective randomized comparative trials of antimicrobial or no therapy in elderly male and female nursing home residents with asymptomatic bacteriuria consistently document no benefit of antimicrobial therapy.** There was no decrease in symptomatic episodes and no improvement in survival. In fact, treatment with antimicrobial therapy increases the occurrence of adverse drug effects and reinfection with resistant organisms and increases the cost of treatment. Therefore asymptomatic bacteriuria in elderly residents of long-term care facilities should not be treated with antimicrobial agents.
41. **f. None of the above.** The American Heart Association's recommendations on the prevention of bacterial endocarditis are based on the patient's risk of developing endocarditis and the likelihood that a procedure will cause bacteremia with an organism that can cause endocarditis. Antibiotic **prophylaxis solely for the prevention of infectious endocarditis is not recommended for GU procedures, even in the setting of high-risk patients including individuals with prosthetic heart valves, previous bacterial endocarditis, cyanotic congenital heart disease, and systemic-pulmonary shunts or conduits.**
42. **d. Catheterization.** Catheter-associated bacteriuria is the most common hospital-acquired infection.

43. **a. Closed drainage.** Careful aseptic insertion of the catheter and maintenance of a closed dependent drainage system are essential to minimize development of bacteriuria.
44. **a. Clean intermittent catheterization (CIC).** Although never rigorously compared with indwelling urethral catheterization, CIC has been shown to decrease lower urinary tract complications by maintaining low intravesical pressure and reducing the incidence of stones.
45. **a. Obesity.** An association between Fournier gangrene and urethral obstruction associated with strictures and extravasation and instrumentation has been well documented. Predisposing factors include diabetes mellitus, local trauma, paraphimosis, periurethral extravasation of urine, perirectal or perianal infections, and surgery such as circumcision or herniorrhaphy.
46. **e. The effects of vaginal estrogen on recurrent UTI risk are related to modulation of local pH and vaginal microflora.** Randomized controlled trials of vaginal estrogen use have not been designed to analyze outcomes such as cancer or cardiovascular disease. However, observational studies have found no increased risk of fracture or breast cancer in women who used vaginal estrogen. Vaginal estrogen is effective in preventing recurrent UTIs in postmenopausal women through lowering the vaginal pH and restoring the normal microenvironment such as lactobacilli. The beneficial effect from vaginal estrogen use can take at least 12 weeks to manifest.
47. **e. Polycystic native kidneys.** Transplant recipients are at higher risk for vesicoureteral reflux. Risk factors for infection include cadaveric graft, diabetes, prolonged hemodialysis prior to transplantation, two episodes of asymptomatic bacteriuria, and female gender.
48. **f. None of the above.** The American Heart Association's recommendations on the prevention of bacterial endocarditis are based on the patient's risk of developing endocarditis and the likelihood that a procedure will cause bacteremia with an organism that can cause endocarditis. Antibiotic prophylaxis solely for the prevention of infectious endocarditis is not recommended for GU

procedures, even in the setting of high-risk patients including individuals with prosthetic heart valves, previous bacterial endocarditis, cyanotic congenital heart disease, and systemic-pulmonary shunts or conduits.

49. **d. Administration of an antimicrobial agent within 60 to 120 minutes of the initiation of a procedure and for a period of time that covers the duration of the procedure.** Surgical antimicrobial prophylaxis entails treatment with an antimicrobial agent before and for a limited time after a procedure to prevent local or systemic postprocedural infections.
50. **a. *Enterococcus*.** Most gram-negative bacteria are capable of producing positive results on a nitrite test. Pseudomonas aeruginosa and most gram-positive organisms do not produce nitrites.
51. **c. History of human papilloma virus (HPV).** HPV history does not impact ability to provide an adequate sample; the other variables are important influencing factors and a catheterized urine sample should be considered in these situations.
52. **a. Family history of postmenopausal UTIs.** Family history of postmenopausal UTIs is least relevant to the patient's immediate evaluation; the other variables list individual behaviors, infection history, and anatomic factors that can have a direct impact on the patient's ongoing UTI diathesis. Family history of UTIs in childhood or young adulthood bears more relevance regarding structural issues (such as vesicoureteral reflux) or genetic risk factors for recurrent infections.

Pathology

1. **c. Is unnecessary.** The figure shows numerous neutrophils within the interstitium and the renal tubules. The neutrophils in the tubules become white blood cell casts. The pathologic findings including an enlarged kidney may persist for several weeks despite appropriate treatment. There is no indication for a biopsy in this patient.

2. **c. Left nephrectomy.** The figure shows the foamy macrophages with neutrophils and cellular debris characteristic of xanthogranulomatous pyelonephritis. It may be associated with renal calculi and *Proteus* infection. *E. coli* is also a common organism found in this disease. Although partial nephrectomy has been performed for a small localized mass in a functioning kidney, a left nephrectomy in this situation is likely required and is necessary to rid the patient of the infection. An associated retroperitoneal inflammatory process with thickening is not uncommon.
3. **d. Treat with a sulfonamide for several months.** Fig. 55.3A shows von Hansemann histiocytes, and Fig. 55.3B demonstrates the Michaelis-Gutmann bodies, both of which are characteristic of malakoplakia. It is thought to be infectious in origin, and therefore the treatment is an extended course of an antibiotic that achieves a high intracellular concentration.

Imaging

1. **d. Right perinephric abscess.** The CT scan is obtained in the late arterial to nephrographic phase of the examination (the aorta is still opacified with contrast agent), before the excretion of the contrast agent. Thus option b is incorrect. There are multiple calculi in the right kidney, which is small and atrophic, indicating a chronic process (thus option a is incorrect). There is thickening of the perinephric fascia, and gas bubbles are seen in the posterior paranephric space, extending to the right flank. In addition, there are fluid collections in the posterior paranephric space and in the soft tissues of the right flank, making option d the most likely diagnosis. Xanthogranulomatous pyelonephritis is a chronic inflammatory condition associated with staghorn calculi. The affected kidney is usually enlarged rather than shrunken, as is the case here (making option e unlikely).
2. **e. Acute pyelonephritis.** The image demonstrates a pelvic kidney with wedge-shaped area of decreased enhancement, characteristic

of acute pyelonephritis. Renal infarcts cause areas of poor perfusion that are more sharply defined and more poorly enhancing than in the present case (making option a unlikely). The clinical history of fever also supports an infection. With renal artery occlusion (option b) the kidney would demonstrate no enhancement. Chronic pyelonephritis causes scarring in the kidney, and the nephrogram is usually normal. The renal contour in the present case is smooth, making option c unlikely. Acute urinary obstruction (option d) is ruled out because the visualized collecting system does not appear dilated.

3. **a. Percutaneous drainage.** The image demonstrates a low-attenuation area in the posterior interpolar region of the left kidney, with perinephric fascial thickening, consistent with a renal abscess. Intravenous antimicrobial therapy with percutaneous drainage of renal abscesses is highly effective and is the treatment of choice. Antimicrobial therapy alone is unlikely to be effective, given the size of the abscess. Nephrectomy, partial nephrectomy, and surgical drainage are rarely indicated in young patients with normally functioning kidneys. Cystoscopy is not warranted.
4. **a. Inpatient antibiotics, Foley catheter placement, and glucose management.** The figure is an example of emphysematous pyelonephritis. The majority (90%) of these patients are treated with medical therapy alone, which consists of antibiotics (often parenteral), bladder drainage, and treatment of comorbid conditions such as poorly controlled diabetes. The need for surgical intervention is rare and is reserved for those cases that respond poorly to initial medical management or severe necrotizing infections.

Chapter Review

1. Bacteriuria and pyuria are not synonymous with a UTI.
2. UTIs are classified based on their presumed site of origin.
3. UTIs can be uncomplicated (occurring in healthy patients with normal urinary tracts) or complicated (associated with factors that

increase the likelihood of bacteriuria and decrease the efficacy of therapy).

4. UTIs are the most common bacterial infection, and, as such, make a significant impact on healthcare costs.
5. The incidence of bacteriuria increases with institutionalization/hospitalization as well as with pregnancy and certain comorbidities that alter lower urinary tract function or cause immunosuppression.
6. To date, no clear association has been described between recurrent uncomplicated UTIs and renal sequelae such as scarring, hypertension, or progressive renal insufficiency.
7. Most UTIs are caused by bacteria, usually originating from the bowel flora.
8. Bacterial virulence factors, including adhesin, play a role in determining which bacteria invade and the extent of infection.
9. Increased epithelial cell receptivity predisposes patients to recurrent UTIs and is a genotypic trait.
10. Obstruction to urine flow is a key factor in increasing host susceptibility to UTIs.
11. Clinical presentation is critically important in considering diagnosis; results of urine testing cannot be analyzed without knowledge of signs and symptoms.
12. Urine must be collected in a manner that minimizes contamination.
13. Formal urinalysis is preferred over dipstick testing.
14. Urine culture results provide bacterial sensitivities.
15. Urine is not sterile.
16. The female urinary microbiome has been identified using RNA sequencing techniques, and knowledge of these microbiota should inform UTI management and interpretation of novel culture techniques such as the expanded quantitative urine culture (EQUC) protocol.
17. The prevalence of asymptomatic bacteriuria varies with age, sex, and comorbid conditions.
18. Untreated asymptomatic bacteriuria is not associated with hypertension or renal insufficiency.

19. Several guidelines recommend not screening for or treating asymptomatic bacteriuria except in specific patient populations.
20. Treatment of asymptomatic bacteriuria contributes to development of multi-drug resistant symptomatic UTIs.
21. Antimicrobial resistance is increasing because of excessive utilization.
22. Antimicrobial selection should be influenced by efficacy, safety, cost, and compliance.
23. Both the choice of the agent as well as the duration of therapy are critical in preventing the perpetuation of antimicrobial resistance as well as adverse events related to treatment.
24. Obtaining a thorough history, with emphasis on prior symptoms, urinalysis and culture results, and triggers for infections, is critical.
25. Sources of possible bacterial persistence must be identified and eradicated.
26. Imaging and cystoscopy are NOT recommended for recurrent uncomplicated UTIs.
27. Prevention of recurrence should focus on non-antibiotic interventions.
28. Acute pyelonephritis classically presents as the abrupt onset of chills, fever, and flank or costovertebral angle tenderness but can present as symptoms as mild as cystitis or as severe as sepsis.
29. Emphysematous pyelonephritis is a life-threatening infection diagnosed radiographically by the presence of gas in the parenchyma or collecting system and can be managed via percutaneous drainage or surgically.
30. Renal abscesses are well delineated by CT and are classically managed with IV antimicrobial agents and drainage. Smaller abscesses may be amenable to conservative treatment with medical management.
31. Pyonephrosis is a bacterial infection in a hydronephrotic kidney. Prompt diagnosis is critical; treatment entails intravenous antimicrobial agents and drainage of the obstructed renal unit.
32. XGP is a chronic renal infection that is often found in poorly functioning renal units obstructed secondary to nephrolithiasis. XGP can be mistaken for renal tumors.

33. Malacoplakia is an unusual inflammatory disease thought to result from abnormal macrophage function. Michaelis-Gutmann bodies are lysosomal inclusion bodies that characterize this disease microscopically.
34. Fournier gangrene is necrotizing fasciitis arising from the perineal skin, scrotum, urethra, or rectum.
35. Emergent surgical debridement and broad-spectrum antimicrobial agents are the essentials of treatment of Fournier gangrene.
36. Periurethral abscess can occur secondarily to urethral stricture or catheterization; treatment entails surgical debridement, suprapubic urinary drainage, and antimicrobial agents.
37. Sepsis is a clinical syndrome characterized by extremes of body temperature, heart rate, respiratory rate, and WBC count that occurs in response to an infection.
38. The principles of management of sepsis include resuscitation, supportive care, monitoring, administration of broad-spectrum antimicrobial agents, and drainage or elimination of infection.
39. The surviving sepsis campaign and early goal-directed therapy has been shown to improve outcomes in critically ill patients.
40. Screening for bacteriuria via urine culture should be performed in all pregnant women during the first trimester.
41. The prevalence of bacteriuria does not change with the occurrence of pregnancy; however, unlike in non-pregnant women, spontaneous resolution of bacteriuria in pregnant women is unlikely.
42. All pregnant women with bacteriuria should be treated.
43. Bacteriuria more commonly progresses to acute pyelonephritis during pregnancy.
44. Screening for bacteriuria is not recommended in elderly patients because there is no relationship between asymptomatic bacteriuria and uncomplicated UTIs and deteriorating renal function; asymptomatic bacteriuria should not be treated.
45. Infections of the urinary tract may present as subtle signs, and a high index of suspicion is often required for diagnosis.
46. Careful aseptic insertion of the catheter and maintenance of a closed, dependent drainage system are essential to minimize development of bacteriuria.

47. The development of catheter-associated bacteriuria is inevitable. Only symptomatic catheter-associated UTIs require treatment.
48. The majority of spinal cord–injured patients with bacteriuria are asymptomatic. Only symptomatic patients require therapy.
49. UTI in patients with spinal cord injury commonly presents as fever; flank, back, or abdominal discomfort; leakage between catheterizations; increased spasticity; malaise; lethargy; and/or cloudy, malodorous urine.
50. Urine culture before the initiation of empirical therapy is essential because spinal cord–injured patients often culture diverse flora with a high probability of bacterial resistance.
51. Clean intermittent catheterization places patients with spinal cord injury at the lowest risk for significant long-term urinary tract complications.
52. Antimicrobial prophylaxis entails treatment with an antimicrobial agent before and for a limited time after a procedure to prevent local or systemic postprocedural infections.
53. The type of procedure and competency of the host defenses determine the need for antimicrobial prophylaxis.
54. Special considerations for antimicrobial prophylaxis include patients undergoing TRUS-Bx, those with a risk of endocarditis and bacteriuria, and patients with indwelling orthopedic hardware.

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Inflammatory and Pain Conditions of the Male Genitourinary Tract

Prostatitis and Related Pain Conditions, Orchitis, and Epididymitis

Michel Pontari

Keywords

chronic pelvic pain syndrome; chronic scrotal pain syndrome; epididymitis; orchitis; prostatitis

Questions

1. A man with chronic pelvic pain has a negative urine culture and no evidence of inflammation on expressed prostatic secretions (EPS) or VB3 or in his seminal plasma. What is his category of prostatitis in the NIH classification?
    a. Category I
    b. Category II
    c. Category IIIA
    d. Category IIIB
    e. Category IV
2. A 65-year-old male undergoes a transrectal ultrasound-guided biopsy for an elevated PSA. He has no pelvic pain. Biopsy shows

no cancer, but he does have many areas of prostate inflammation. What is his NIH prostatitis category?

a. Category I
b. Category II
c. Category IIIA
d. Category IIIB
e. Category IV

3. A 58-year-old male with a history of autoimmune pancreatitis presents urinary retention. He is refractory to medical therapy for benign prostatic hyperplasia (BPH), and a transurethral resection of the prostate (TURP) is planned. What type of histology may be expected from the prostate resection?

a. No inflammation
b. Granulomatous prostatitis
c. Scant periglandular mononuclear infiltrate
d. Heavy intraductal lymphocytic infiltrate
e. Immunoglobulin G4 (IgG4)-positive plasma cells and fibrosis

4. What is the most common causative organism in patients with category I acute bacterial prostatitis that develops after lower urinary tract manipulation?

a. *Escherichia coli*
b. *Pseudomonas*
c. *Proteus*
d. *Klebsiella*
e. *Enterococcus*

5. A man presents after transrectal prostate biopsy with a fever of 102°F, rigors, and many white blood cells and bacteria in the urine. Which antibiotic is the best choice for initial therapy until urine and blood culture results are available?

a. Ciprofloxacin
b. Levofloxacin
c. Trimethoprim-sulfamethoxazole
d. Cefpodoxime
e. Meropenem

6. A 35-year-old human immunodeficiency virus (HIV)-positive man presents with fever, dysuria, and leukocytosis. Imaging reveals a

0.9-cm hypoechoic area in the prostate consistent with a prostate abscess. Initial management for this patient is:

a. intravenous (IV) antibiotics
b. prostate massage to help drain the abscess
c. percutaneous aspiration
d. transurethral resection
e. open perineal drainage

7. A 41-year-old male presents with his second urinary tract infection in the past 4 months. He is growing the same *E. coli* as his previous culture. Evaluation should include:

a. computed tomography (CT) urogram
b. uroflow and postvoid residual urine to rule out retention
c. cystoscopy
d. four-glass urine test
e. nuclear amplification test for gonorrhea and chlamydia

8. What is the relationship between inflammation (white blood cells [WBCs]) found in expressed prostatic fluid and symptoms in men with chronic pelvic pain syndrome?

a. Greater WBCs in men with predominantly voiding symptoms plus pelvic pain
b. Inverse correlation between WBCs and symptoms
c. Direct correlation between WBCs and symptoms
d. No correlation between WBCs and symptoms
e. Greater WBCs only in men with prior history of acute prostatitis

9. What psychological factor is correlated with overall pain along with urinary symptoms and depression in men with chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS)?

a. Neuroticism **START**
b. Obsessive compulsive disorder
c. Anger
d. Catastrophizing
e. Insecurity

10. What is the most common overlapping pain condition (COPC) in men with CP/CPPS?

a. Fibromyalgia

b. Chronic fatigue syndrome
c. Irritable bowel syndrome (IBS)
d. Temporomandibular joint disorder
e. Tinnitus

11. In a case-crossover study in the NIH Multidisciplinary Approach to Pelvic Pain (MAPP) study, what correlated with development of symptom flares?
    a. Psychological stress
    b. Caffeine
    c. Alcohol
    d. Recent sexual activity
    e. Physical activity
12. Compared with age-matched controls, men with CP/CPPS have a greater prevalence of which of the following?
    a. BPH
    b. Urinary incontinence
    c. Premature ejaculation
    d. Condyloma
    e. Phimosis
13. The NIH–Chronic Prostatitis Symptom Index (NIH-CPSI) score has been used in many trials as an outcome measure for treatment of CP/CPPS. It is also used for diagnosis and epidemiologic studies. The Genitourinary Problem Index (GUPI) is a modified version that better reflects what symptoms found in men with CP/CPPS but not part of the NIH-CPSI?
    a. Penile pain
    b. Epididymal pain
    c. Scrotal skin pain
    d. Back pain
    e. Pain with bladder filling
14. Which of the following is an optional study in the evaluation of men with CP/CPPS?
    a. Symptom assessment
    b. Midstream urine for urinalysis and culture
    c. Abdominal exam
    d. Examination of pelvic floor tender points

e. Cystoscopy

15. On electromyography (EMG) study during urodynamics, a man with pelvic pain, poor stream, and pain with sitting has retained electrical activity of the external sphincter during voiding. He is grossly neurologically intact. What is his diagnosis?
    a. Detrusor underactivity
    b. Upper motor neuron lesion
    c. Pseudodyssynergy
    d. Urethralis fugax
    e. Detrusor overactivity
16. In meta-analyses of treatments for CP/CPPS, which combinations of therapy show the best results?
    a. Antiinflammatory medications and α-blockers
    b. Antibiotics and α-blockers
    c. 5α-reductase inhibitors and α-blockers
    d. Skeletal muscle relaxants and antiinflammatory medications
    e. Antibiotics and 5α-reductase inhibitors
17. A 38-year-old male presents with a 4-month history of pelvic pain. Urine cultures are negative. He has some generalized increase in anxiety. On exam, he has significant increase in tone of the pelvic floor muscles around the prostate, especially the obturator internus. Management should consist of stress reduction and which of the following?
    a. Referral to a psychiatrist
    b. Oral valium
    c. Microwave thermotherapy
    d. Pelvic floor physical therapy
    e. Injection of onabotulinumtoxinA into the perineal body
18. A 65-year-old male undergoes prostate biopsy for an elevated PSA of 5.4. Pathology reveals extensive glandular and periglandular inflammation with both acute and chronic inflammation. The best next step is:
    a. 4-week course of antiinflammatory medications.
    b. start finasteride.
    c. 4-week course of antibiotics and then reassess.
    d. observation.

e. repeat biopsy.

19. UPOINT is a phenotyping tool that captures multiple aspects of symptoms and finding in men with CPPS. What does the "N" stand for in UPOINT?
    a. Nonbacterial
    b. Noninflammatory
    c. Neurologic
    d. Normal prostate exam
    e. Normotensive
20. Autoimmune orchitis is characterized by the presence of:
    a. antisperm antibodies (ASAs)
    b. antisperm granulomas
    c. testicular microinfarcts
    d. microlithiasis
    e. antilipid antibodies
21. A 25-year-old male presents with the acute onset of left epididymal pain. The left epididymis is tender to palpation and swollen. There are 1 to 3 WBCs per high-power field on urinalysis. He is afebrile. He denies being sexually active. The most likely cause for his acute epididymitis is:
    a. *E. coli*
    b. Pseudomonas
    c. Proteus
    d. Brucellosis
    e. Chlamydia
22. A 39-year-old male has had chronic orchialgia for 1 year. He has failed a course of antibiotics and antiinflammatory medications. He has no other medical conditions and no back pain. On examination his right testis is mildly tender. On digital rectal exam he has minimal prostate tenderness but has tenderness to the right obturator internus muscle. The next best step in treatment is:
    a. gabapentin
    b. pregabalin
    c. pelvic floor physical therapy
    d. microsurgical denervation of the right spermatic cord
    e. referral to pain management

23. Which area is part of the so-called trifecta nerve complex that should be addressed in microsurgical denervation of the spermatic cord for chronic scrotal pain?
    a. Epididymal tail
    b. Cremasteric muscle fibers
    c. Rete testis
    d. Epididymal tail
    e. Testicular artery
24. What technique has been noted to increase the success rate of epididymectomy performed for post vasectomy pain syndrome?
    a. Ligation of surrounding lymphatics
    b. Ligation of the spermatic veins
    c. Preservation of surrounding lymphatics
    d. Incorporation of the distal portion of the vasectomy site
    e. Inguinal approach
25. What technique has been reported to increase the success rate of orchiectomy for chronic scrotal pain?
    a. Ligation of surrounding lymphatics
    b. Ligation of the spermatic veins
    c. Preservation of surrounding lymphatics
    d. Scrotal approach
    e. Inguinal approach

Answers

1. **d. Category IIIB.** TheNIH classification is IIIB, chronic prostatitis chronic pelvic pain syndrome. He has no infection to suggest acute or chronic bacterial prostatitis, and he is symptomatic and therefore not category IV. The absence of inflammatory cells separates the subclasses of IIA and IIIB. He has no inflammation on EPS, VB3, or seminal plasma and therefore is category IIIB.
2. **e. Category IV.** Category IV is asymptomatic inflammatory prostatitis. Asymptomatic means no symptoms of pelvic pain. This usually diagnosed on specimens taken for treatment of other problems such as BPH, prostate cancer, or infertility.

3. **e. Immunoglobulin G4 (IgG4)-positive plasma cells and fibrosis.** IgG4-positive plasma cells and a characteristic pattern of fibrosis are seen in IgG4 disease. First described in autoimmune pancreatitis, multiple previously described disorders including Ormond disease (retroperitoneal fibrosis) are currently considered to fall within this category. One of the characteristics of IgG4 disease is a dramatic response to corticosteroids.
4. **a. *Escherichia coli.*** Although *E. coli* is still the most commonly found organism in patients with acute bacterial prostatitis (ABP) after lower urinary tract manipulation, *Pseudomonas* is far more common in cases from manipulation compared with spontaneous or after transrectal prostate biopsy.
5. **e. Meropenem.** The Update of the American Urological Association White Paper on the prevention and treatment of complications after prostate biopsy recommends that patients who present with a fever after prostate biopsy should not be offered fluoroquinolones or trimethoprim-sulfamethoxazole. Management should consist of aggressive resuscitation and broad-spectrum antibiotic coverage—carbapenems, amikacin, or second- and third-generation cephalosporin (after urine and blood cultures).
6. **a. Intravenous (IV) antibiotics.** In men with prostate abscess, smaller lesions less than 1 to 2 cm can be treated conservatively with antibiotics. Larger lesions or lesions that do not respond to antibiotics can be treated with percutaneous aspiration or transurethral unroofing.
7. **b. Uroflow and postvoid residual urine to rule out retention.** In a man younger than 45 years, imaging is not usually helpful. A cystoscopy could be done if he has an elevated residual to rule out stricture if he has a decreased uroflow and elevated postvoid residual urine. The four-glass test is no longer routinely used for cases of suspected chronic bacterial prostatitis. The nuclear amplification test would be indicated if he had a urethral discharge.
8. **d. No correlation between WBCs and symptoms.** No correlation between WBCs and symptoms. In the Chronic Prostatitis Collaborative Research Network (CPCRN) study, men with no

symptoms had equal amounts of inflammation in EPS as those with pelvic pain.

9. **d. Catastrophizing.** Helplessness and catastrophizing predict overall pain along with urinary symptoms and depression in men with CP/CPPS.
10. **c. Irritable bowel syndrome (IBS).** The most COPC is IBS. Men should be queried about signs and symptoms of bowel dysfunction such as constipation, diarrhea, and the relationship to pelvic pain to bowel movements. Referral to a gastroenterologist should be made if indicated.
11. **d. Recent sexual activity.** Only recent sexual activity and urinary tract infection (UTI)-like symptoms were predictive of flare. Items that did not predict a flare included diet, physical activity or sedentary behavior, stress, and constipation.
12. **c. Premature ejaculation.** Men with CP/CPPS are more likely to have ejaculatory pain and premature ejaculation.
13. **e. Pain with bladder filling.** The GUPI is a modified version that contains questions about bladder pain. These symptoms were found in approximately 40% of men in the NIH-sponsored MAPP study.
14. **e. Cystoscopy.** Cystoscopy is an optional study. Cystoscopy can be used in men with decreased uroflow and/or elevated postvoid residual urine to evaluate for urethral stricture. Cystoscopy should also be used in men who give a history of pain with bladder emptying and/or relieved by bladder emptying, suggestive of interstitial cystitis/bladder pain syndrome (IC/BPS) because in a small set of these patients, a Hunner ulcer is present which responds to fulguration or intralesional injections.
15. **c. Pseudodyssynergy.** Pseudodyssynergy is retained electrical activity of the external sphincter during voiding in the absence of abdominal straining, and/or brief and intermittent closing of the membranous urethra during voiding detected by electromyography, or fluoroscopy in videourodynamics. These men are considered to have pelvic floor dysfunction.
16. **b. Antibiotics and α-blockers.** Meta-analyses have shown the best response for a combination of antibiotics and α-blockers.

17. **d. Pelvic floor physical therapy.** Although he has anxiety, the best treatment is addressing the increased pelvic floor tension by referral a physical therapist for pelvic floor physical therapy (PT).
18. **d. Observation.** No treatment for asymptomatic prostate inflammation (category IV) is necessary. Treatment with antibiotics is not indicated and should not be used to "treat" his elevated PSA to try to get it back to normal levels. He should be followed for his PSA elevation and have periodic repeat PSA and digital rectal examinations.
19. **c. Neurologic.** The N stands for Neurologic. This is for men with widespread pain conditions, with pain outside the pelvis, and overlapping conditions such as IBS, fibromyalgia, or chronic fatigue syndrome. They are thought to possibly have upregulation of the sensitivity of their central nervous system (centralization or central sensitization) and may respond to medications that treat neuropathic pain.
20. **a. Antisperm antibodies (ASAs).** Autoimmune orchitis is defined as an autoimmune aggression of the testes characterized by the presence of ASAs. Primary autoimmune orchitis is primarily a problem in the evaluation of male infertility. The main causes of secondary autoimmune orchitis are usually associated with a primary vasculitis, particularly Behçet disease, polyarteritis nodosa, and Henoch-Schönlein purpura.
21. **e. Chlamydia.** Sexually transmitted diseases and particularly chlamydia are the most common cause of acute epididymitis. Patient should be suspected of chlamydia even if they deny recent intercourse, because the infection does not always correlate with the reported sexual history.
22. **c. Pelvic floor physical therapy.** He has signs or history to suggest a centralized pain condition. Pelvic floor therapy is emerging as a treatment for chronic orchialgia. All conservative measures should be tried prior to undergoing surgery.
23. **b. Cremasteric muscle fibers.** Three areas identified by Parekatil et al. that had the highest areas of wallerian degeneration (WD) were the cremasteric muscle fibers, the perivasal sheath, and the posterior lipomatous tissue; these areas were called the trifecta

nerve complex and is thought to form the anatomic basis for relief from the denervation procedure.

24. **d. Incorporation of the distal portion of the vasectomy site.** Better rates are reported in cases with incorporation of the distal portion of the vasectomy site with the epididymectomy specimen.
25. **e. Inguinal approach.** Better results have been reported with the inguinal approach to orchiectomy performed for chronic scrotal pain.

Chapter Review

1. Men with category I acute prostatitis present with acute symptoms of dysuria and frequency and may have fever.
2. The most common causative organism is *E. coli.*
3. Types of bacteria and antibiotic sensitivities differ by etiology: spontaneous, after transurethral procedure, or after transrectal biopsy.
4. Men with acute prostatitis after transrectal biopsy should be treated with carbapenems, amikacin, or send and third-generation cephalosporins.
5. A suprapubic tube should be used if long-term drainage is needed.
6. Failure to respond to therapy should prompt imaging to look for a prostatic abscess.
7. Chronic bacterial prostatitis is characterized by recurrent urinary tract infections with the same organism
8. Unlike men with category III prostatitis/chronic pelvic pain syndrome, they are then relatively asymptomatic between episodes.
9. Urinary retention as a cause of recurrent UTI must be ruled out.
10. Men younger than 45 years do not need imaging but need assessment for a urethral stricture.
11. Treatment is with antibiotics that achieve adequate prostate concentrations
12. For refractory cases, daily suppressive antibiotics or TURP can be used.

13. Men with CP/CPPS have equal amounts of bacteria in urine, prostate fluid, and seminal plasma as men without pain.
14. There is evidence of central sensitization of the CNS in both afferent and efferent nerves.
15. In those with inflammation in VB3, EPS, or seminal plasma, the degree of inflammation does not correlate with symptoms.
16. Helplessness and catastrophizing predict overall pain along with urinary symptoms and depression.
17. Men with CPPS also have a lower baseline adrenocorticotropic hormone (ACTH) level and blunted ACTH rise in response to stress than men without symptoms.
18. Men CP/CPPS have increased pain sensitivity compared with controls.
19. CP/CPPS is a diagnosis of exclusion and should be made only after a thorough search for other causes of pelvic pain.
20. CP/CPPS has potential issues outside of the pelvis (and usual scope of urology) such as psychological issues and neurologic problems.
21. Patients with CP/CPPS must be assessed for other chronic pain syndrome such as IBS, fibromyalgia, and chronic fatigue syndrome.
22. Patients with tenderness on pelvis exam may benefit from pelvic floor physical therapy/myofascial release.
23. Therapy for CPPS is best done in a multimodal fashion; the symptom classification UPOINT can be helpful in directing evaluation and subsequently therapy.
24. Repeated courses of antibiotics in the absence of a positive urine culture are not indicated.
25. The best treatment combination from meta-analyses is antibiotics plus α-blockers.
26. Men with bladder pain should be considered for bladder-specific therapies.
27. Men with tenderness on pelvic floor exam should be referred for pelvic floor physical therapy.
28. Sacral neuromodulation can be used in men with pain who also have frequency and urgency.
29. Treatment is best undertaken as simultaneous multimodal therapy addressing all aspects of the patient’s symptoms.

30. Viral orchitis is most commonly caused by the mumps virus.
31. For scrotal pain a duplex Doppler ultrasound should be performed to assess for anatomic or congenital abnormality, torsion, or mass.
32. Pelvic floor physical therapy should be considered in men with orchialgia and abnormal tone on digital rectal exam.
33. Acute epididymitis is more commonly caused by an STD in men younger than 35 years and enteric organism in men older than 35 years.
34. For microsurgical spermatic cord denervation (MSDC) three areas that should be addressed are the cremasteric muscle fibers, the perivasal sheath, and the posterior lipomatous tissue.
35. The distal vas deferens should be incorporated into the surgical specimen during an epididymectomy for postvasectomy pain syndrome.

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Interstitial Cystitis/Bladder Pain Syndrome and Related Disorders

Robert M. Moldwin, and Philip M. Hanno

Keywords

bladder pain syndrome; bladder wall; interstitial cystitis; lower urinary tract

Questions

1. Essential for the diagnosis of interstitial cystitis/bladder pain syndrome (IC/BPS) is the presence of:
    a. urinary urgency.
    b. pain or discomfort related to the bladder.
    c. glomerulations on cystoscopy.
    d. Hunner lesion (HL).
    e. urinary frequency.
2. The definition of IC proposed by the National Institute of Arthritis, Diabetes, Digestive and Kidney Diseases (NIDDK) is best considered a:
    a. de facto definition of the disease.
    b. diagnostic pathway.
    c. definition applicable mainly to clinical research studies.
    d. historic document of no current value.
    e. purely symptom-based description of IC/BPS.

3. A clinical factor that is more common in IC/BPS patients with HLs than in non-HL patients is:
    a. IC/BPS patients with HL typically present with a history of microscopic hematuria.
    b. IC/BPS patients with HL have a higher mean age.
    c. IC/BPS patients with HL present with higher visual analog scores (VAS) for pain than wnon-HL patients.
    d. Treatment pathways are identical for IC/BPS patients regardless of the presence or absence of HLs.
    e. IC/BPS patients with and without HLs are indistinguishable upon office cystoscopy.
4. IC/BPS symptom and problem indices have been validated to:
    a. monitor disease progression or regression with or without treatment.
    b. correctly choose who should undergo cystectomy and diversion.
    c. determine on whom to perform diagnostic testing.
    d. accurately diagnose IC/BPS.
    e. determine appropriate candidates for clinical research.
5. Which of the following statements best categorizes the natural history of IC/BPS?
    a. The onset is generally insidious and progressive, occurring gradually over many years.
    b. Loss of bladder capacity over time is the rule.
    c. Symptoms follow a culture-documented urinary tract infection.
    d. Symptom resolution regardless of treatment after 1 to 2 years.
    e. Although clinical presentation is usually subacute, many patients have prodromal symptoms that occur decades before a formal diagnosis.
6. Which statement best describes the relationship of IC/BPS to bladder cancer?
    a. IC/BPS is a premalignant lesion.
    b. IC/BPS is often associated with bladder cancer.
    c. A positive urine cytology can safely be ignored in patients with IC/BPS.

d. The vast majority of reports fail to document an association of IC/BPS with subsequent development of bladder cancer.
e. Dysplasia is a typical pathologic finding on bladder biopsy in IC/BPS patients.

7. American Urological Guidelines require the following for a diagnosis of IC/BPS:
   a. Pain or discomfort perceived to be related to the urinary bladder
   b. Age >18
   c. Urinary urgency
   d. Pelvic discomfort that worsens with bladder filling
   e. Presence of petechial submucosal hemorrhages upon short-duration low pressure (60 to 80 cm H_2O) hydrodistention
8. The antibiotic of choice for diagnosed IC/BPS is:
   a. doxycycline.
   b. none.
   c. gentamicin.
   d. ciprofloxacin.
   e. amoxicillin.
9. The cell *most likely* to play a central role in the pathogenesis of IC/BPS without HLs is the:
   a. granulocyte.
   b. lymphocyte.
   c. mast cell.
   d. platelet.
   e. eosinophil.
10. Which statement best categorizes the potassium chloride test?
   a. It is soothing and calming to the painful bladder.
   b. It has high sensitivity and specificity for diagnosing IC/BPS.
   c. It is an important element in choosing effective therapy.
   d. It provides proof of abnormal mucosal permeability.
   e. None of the above.
11. A circumscribed inflammatory bladder lesion:
   a. is required to make a diagnosis of IC/BPS.
   b. is generally found in less than 30% of IC/BPS patients.

c. was not considered a part of the syndrome when it was initially described by Hunner.
d. is synonymous with glomerulation.
e. is pathognomonic of IC/BPS even in the absence of symptoms.

12. Exclusive use of the NIDDK criteria to diagnose IC/BPS would result in:
    a. an accurate depiction of the true prevalence of the condition.
    b. an improved treatment algorithm.
    c. increased diagnostic specificity.
    d. increased diagnostic sensitivity.
    e. a minimum of diagnostic testing and significant cost savings.
13. All but which of the following disorders have a much higher prevalence in the IC/BPS population than in the general population?
    a. Irritable bowel syndrome
    b. Diabetes
    c. Fibromyalgia
    d. Allergy
    e. Chronic fatigue syndrome
14. A 27-year-old female presents to you with a history consistent with IC/BPS, but she also reports gross hematuria, and cystoscopic exam shows severe inflammation of the bladder wall. Further evaluation needed includes:
    a. hydrodistention of bladder with fulguration of inflammatory disease.
    b. laparoscopy for diagnosis of endometrial implants.
    c. a history of recreational drug use.
    d. bladder biopsy for mast cell count.
    e. HgbA1C.
15. A 45-year-old female IC/BPS patient presents with deep dyspareunia, constipation dominant IBS, urinary hesitancy, and the need to strain with urination. Which therapy is the LEAST likely to be helpful?
    a. Physical therapy
    b. Behavioral modification

c. Neuromodulation
d. Chemodeinnervation of the bladder
e. Skeletal muscle relaxants

16. The central role of histopathology in IC/BPS is to:
    a. determine whether or not the patient has an HL.
    b. help determine the most efficacious treatment modality.
    c. predict prognosis.
    d. rule out other disorders that might be responsible for the symptoms.
    e. confirm the diagnosis with pathologic criteria.
17. Which of the following has the least in common with IC/BPS?
    a. Vulvodynia
    b. Chronic bacterial prostatitis
    c. Orchialgia
    d. Penile pain
    e. Perineal and scrotal pain
18. Urodynamic findings typical of IC/BPS include:
    a. uninhibited detrusor contractions.
    b. obstructed flow patterns.
    c. abnormal bladder compliance.
    d. decreased capacity and hypersensitivity.
    e. increased volume at first urge to void.
19. The finding of glomerulations:
    a. should modify treatment only when cystoscopy is performed with the patient under anesthesia.
    b. is of no significance in an asymptomatic patient.
    c. indicates a likelihood of response to laser fulguration of the bladder.
    d. is present only in patients with IC/BPS.
    e. is sufficient to make a diagnosis of IC/BPS.
20. Amitriptyline's potential beneficial effects for the IC/BPS patient include all of the below EXCEPT:
    a. its ability to promote sleep
    b. its anticholinergic effect
    c. its antihistaminic effect
    d. its effect on neuropathic pain

e. its antidepressant effect

21. Which test is potentially most helpful for prognosis and therapy?
    a. Potassium chloride test
    b. Intravesical heparin trial
    c. Cystoscopy and low-pressure bladder hydrodistention
    d. Bladder biopsy
    e. Urodynamics
22. Which of the following treatments is targeted to the glycosaminoglycan layer of the bladder?
    a. Sodium pentosan polysulfate
    b. Amitriptyline
    c. Hydroxyzine
    d. l-Arginine
    e. None of the above
23. Which of the following intravesical treatments has shown proven efficacy for IC/BPS in pivotal U.S. Food and Drug Administration trials?
    a. BCG (bacille Calmette-Guérin)
    b. 4% lidocaine
    c. Botulinum toxin
    d. Heparin
    e. None of the above
24. Which of the following statements is true of opioids in the management of the IC/BPS patient?
    a. They have no place in the treatment of a chronic, nonmalignant condition such as IC/BPS.
    b. Chronic use will usually produce physical dependency.
    c. They generally result in drug addiction when used for chronic pain.
    d. They tend to cause diarrhea and sleeplessness.
    e. All of the above.
25. Which of the following is a reasonable surgical procedure to relieve the pain of IC/BPS?
    a. Transurethral fulguration of HL
    b. Reduction cystoplasty
    c. Sympathectomy and intraspinal alcohol injections

d. Cystolysis
e. Transvesical infiltration of the pelvic plexuses with phenol

26. The most important early step in the management of IC/BPS is:
    a. initiating intravesical treatment.
    b. patient education.
    c. starting oral pentosan polysulfate therapy.
    d. physical therapy.
    e. strict adherence to "IC" diet.
27. A finding of detrusor overactivity on urodynamics in a patient with bladder pain in the absence of urinary urgency indicates:
    a. the patient needs treatment with antimuscarinic medication.
    b. the patient does not have IC/BPS.
    c. a urinary tract infection is likely.
    d. neuromodulation would be the most effective treatment.
    e. none of the above.
28. Men with irritative voiding symptoms and pelvic pain should be evaluated for:
    a. bacterial cystitis
    b. bacterial prostatitis.
    c. BPS.
    d. bladder carcinoma in situ.
    e. all of the above.
29. An IC/BPS patient with HLs, unremitting pain, and an anesthetic bladder capacity of 75 cc wishes to proceed with supratrigonal cystectomy with bladder augmentation. The most important factor to consider before embarking upon this procedure is:
    a. urethral pain
    b. body image
    c. preoperative bowel function
    d. the presence of HLs on the trigone
    e. the presence of extravesical fat stranding
30. Emotional, sexual, or physical abuse can be categorized as:
    a. risk factors for BPS.
    b. behaviors often attributed to patients with IC/BPS.
    c. unequivocally unrelated to IC/BPS.

d. rare adverse events caused by medications used to treat IC/BPS.
e. conditions for which there are no data to allow any tentative conclusions with regard to the relationship to IC/BPS.

31. The only phenotype of IC/BPS currently shown to have a unique response to therapy and different natural history is:
    a. nocturia.
    b. daytime frequency.
    c. absence of bladder pain.
    d. glomerulation.
    e. HL.
32. Intravesical therapy for the IC/BPS patient will probably be LEAST effective to reduce pelvic pain in:
    a. the patient with provoked vestibulodynia and high-tone pelvic floor dysfunction.
    b. the patient with fibromyalgia.
    c. the patient with an HL.
    d. the patient with diabetic neuropathy.
    e. the patient with a small-capacity bladder.

Answers

1. **b. Pain or discomfort related to the bladder.** Pain, pressure, or discomfort related to the bladder is necessary to make a diagnosis of BPS. Urgency, frequency, and the presence of glomerulations or Hunner ulcer on endoscopy are often associated with IC/BPS, but the presence of pain or discomfort is the primary component.
2. **c. Definition applicable mainly to clinical research studies.** The definition of IC proposed by the NIDDK is best considered a definition applicable for use in research studies. It was never meant to define the disease but rather was developed to ensure that patients included in basic and clinical research studies were homogeneous enough that experts could agree on the diagnosis.
3. **b. IC/BPS patients with HL have a higher mean age.** Otherwise, the two patient groups share remarkable similarities in clinical

presentation. Treatment algorithms for the patient with HLs are similar; however, these HL patients are more likely to respond to "bladder-centric" interventions such as lesion fulguration.

4. **a. Monitor disease progression or regression with or without treatment.** Symptom and problem indices like the one developed by O'Leary and Sant are not intended to diagnose IC/BPS. Like the American Urologic Association Symptom Score for benign prostatic hypertrophy, these indices are designed to evaluate the severity of symptoms and to monitor disease progression or regression and response to treatment.
5. **e. Although clinical presentation is usually subacute, many patients have prodromal symptoms that occur decades before a formal diagnosis.** The subacute presentation of IC/BPS is one factor that often prompts empiric therapy for overactive bladder syndrome or bacterial cystitis. Further questioning often reveals LUTS or pelvic pain/discomfort that dates back decades or to childhood.
6. **d. The vast majority of reports fail to document an association of IC/BPS with subsequent development of bladder cancer.** Until recently, no relationship has ever been shown between BPS and the subsequent development of bladder carcinoma. In the 1970s, the Mayo Clinic documented bladder cancer in 12 of 53 men who had been treated for IC, but the association was the result of incorrect diagnosis rather than progression. Peters and others have noted that patients with bladder cancer can be misdiagnosed with IC/BPS. A study from Taiwan reported a 2.95 relative risk compared with controls.
7. **a. Pain or discomfort perceived to be related to the urinary bladder.** Pain or discomfort is the central symptom of IC/BPS and is what usually drives the urinary frequency. Although patients often describe pelvic discomfort with bladder filling, it is not an association that is needed for current diagnosis. Urinary urgency is a term that is frequently misinterpreted by patients (painful urgency or urgency related to fear of urine loss) and not necessary for diagnosis. Bladder hydrodistention, although possibly useful as a form of therapy, has not shown significant sensitivity and

specificity for an IC/BPS diagnosis. On the other hand, factors such as the anesthetic bladder capacity measured at the time of hydrodistention may have prognostic significance.

8. **b. None.** Antibiotics are not indicated for the treatment of IC/BPS, nor have they been implicated as a causative factor. An empiric trial of doxycycline is reasonable in patients who have never had an antibiotic trial to treat the symptoms. Numerous studies have concluded that it is unlikely that active infection is involved in the ongoing pathologic process or that antibiotics have a role to play in treatment.
9. **c. Mast cell.** Mast cells are strategically localized in the urinary bladder close to blood vessels, lymphatics, nerves, and detrusor smooth muscle. **IC/BPS appears to be a syndrome with neural, immune, and endocrine components in which activated mast cells may play a central, although not primary, role in many patients.**
10. **e. None of the above.** As many as 25% of patients who meet the NIDDK criteria for IC/BPS will have a negative KCl test. It is positive in the majority of patients with radiation cystitis, urinary tract infection, or nonbacterial prostatitis and in women with pelvic pain. It is neither sensitive nor specific for IC/BPS, is uncomfortable for patients, and does not help to guide therapeutic decisions.
11. **b. Is generally found in less than 30% of IC/BPS patients. The HL (formerly termed "Hunner ulcer")** is found in a minority of patients with symptoms of IC/BPS. Prevalence of these lesions may vary based upon the method to diagnose. The finding can usually be easily identified upon office cystoscopy. Some authorities suggest that hydrodistention is necessary to better identify the vulnerable mucosa. Most recently, narrow-band imaging has been used to better identify these focal regions.
12. **c. Increased diagnostic specificity.** Exclusive use of the NIDDK criteria to diagnose BPS would result in increased specificity and decreased sensitivity. Ninety percent of expert clinicians in the NIDDK database study agreed that patients diagnosed with IC by those criteria had IC. However, 60% of patients diagnosed by these

clinicians as having IC/BPS did not fulfill the NIDDK criteria. Using the criteria as a basis for diagnosis would probably exclude the majority of patients with this symptom complex from the correct diagnosis.

13. **b. Diabetes. Fibromyalgia, irritable bowel syndrome, chronic fatigue syndrome, and atopic allergic reactions are overrepresented in the IC/BPS population.** Studies are ongoing to find out the reason for such relationships in the NIDDK MAPP (Multidisciplinary Approach to the Study of Chronic Pelvic Pain) Network.
14. **c. A history of recreational drug use.** Ketamine cystitis may present with signs and symptoms identical to IC/BPS and is often associated with lower and upper tract inflammatory disease. After excluding other pathologies, that is, bladder cancer, tuberculosis cystitis, and without knowing this past history, the clinician could easily move forward with IC/BPS-based therapies when other approaches would be indicated.
15. **d. Chemodenervation of the bladder.** Off-label use of botulinum toxin A (BTX-A) in the IC/BPS may be considered when more conservative therapies fail. The patient described has symptoms suggestive of high-tone pelvic floor dysfunction with some obstructive voiding symptoms. Physical examination of this patient might reveal hypertonicity of the pelvic floor musculature along with discrete muscular tender points. Caution should be exercised when considering the use of (BTX-A) in patients who complain of obstructive voiding difficulties.
16. **d. Rule out other disorders that might be responsible for the symptoms.** The primary value of histopathology in IC/BPS is to rule out other diseases that may account for the symptoms and would be most benefit if inflammatory disease is identified. There is no pathognomonic histologic finding for IC/BPS, nor can histology predict prognosis. Even a severely abnormal microscopic picture does not necessarily indicate a poor prognosis. At this time, no data suggest that the treatment algorithm can be rationally predicated on the basis of the histologic findings alone.

17. **b. Chronic bacterial prostatitis.** IC/BPS can be considered one of the pain syndromes of the urogenital and rectal area, all of which are well described but poorly understood. These include vulvodynia, orchialgia, perineal pain, penile pain, and rectal pain. Bacterial prostatitis is a well-understood entity with a known etiology and generally responds to treatment directed at the offending organism. NIH type 1 includes acute bacterial prostatitis and NIH type 2 denotes chronic bacterial prostatitis. Unlike NIH type 3 chronic pelvic pain syndrome/nonbacterial prostatitis, types 1 and 2 have no relationship to IC/BPS.
18. **d. Decreased capacity and hypersensitivity.** Cystometry in conscious IC/BPS patients generally demonstrates normal function, the exception being decreased bladder capacity and hypersensitivity. Pain on bladder filling, which reproduces the patient's symptoms, is very suggestive of IC. Bladder compliance in patients with IC is normal, as hypersensitivity would prevent the bladder from filling to the point of noncompliance.
19. **b. Is of no significance in an asymptomatic patient.** Glomerulations are not specific for IC/BPS, and seem to have little diagnostic or therapeutic value. Glomerulations can be seen after radiation therapy, in patients with bladder carcinoma, after exposure to toxic chemicals or chemotherapeutic agents, and in patients undergoing dialysis or after urinary diversion when the bladder has not filled for extended periods.
20. **e. Its antidepressant effect. Amitriptyline may be of clinical benefit for all of its qualities listed a to d.** Although the anticholinergic effect may improve symptoms in those patients with concomitant overactive bladder syndrome, negative effects may include a worsening of constipation and/or worsening of obstructive voiding symptoms. The doses to achieve clinical benefit for IC/BPS are generally lower than those used to treat major depressive illness.
21. **c. Cystoscopy and low-pressure bladder hydrodistention.** Bladder hydrodistention with the patient under anesthesia is a common therapeutic modality used for IC/BPS, frequently as part of the diagnostic evaluation. Its primary value is in diagnosis of an

HL. Between 30% and 50% of patients experience some short-term relief in symptoms after the procedure. If an HL is present, therapeutic response to resection or fulguration is excellent in many patients. About 30% of patients will note a brief exacerbation in their symptoms following hydrodistention. A bladder capacity under anesthesia of less than 200 mL is a sign of poor prognosis.

22. **a. Sodium pentosan polysulfate.** The target of sodium pentosan polysulfate therapy is the glycosaminoglycan (GAG) layer of the urothelium. This agent is an oral analog of heparin. The proposed mechanism of action is the correction of a GAG dysfunction, thus presumably reversing the abnormal epithelial permeability. Chronic use has recently been associated with pigmentary maculopathy and visual disturbances in some patients.
23. **e. None of the above.** None of these treatments has been proven efficacious for IC/BPS in double-blind placebo-controlled trials.
24. **b. Chronic use will usually produce physical dependency.** Unlike other classes of analgesics, they have no therapeutic ceiling, dosing being limited by tolerance of side effects. They tend to cause constipation and can cause sedation. Opioids can be very useful in a subset of IC/BPS patients with severe disease and in the setting of "universal precautions."
25. **a. Transurethral fulguration of HL.** Transurethral fulguration and/or steroid injection of an HL can provide symptomatic relief. None of the other procedures listed have any place in the treatment of IC/BPS.
26. **b. Patient education.** Patient education is the most important step in the initial treatment of IC/BPS. This condition is chronic, the symptoms wax and wane, and remissions are not uncommon. It lends itself to practitioner abuse, and the uninformed, desperate patient is easy prey. Treatment is symptom-driven, and an informed patient makes the best decisions.
27. **e. None of the above.** Clinically insignificant detrusor overactivity may be seen in 15% of patients with IC/BPS, a rate of involuntary contractions that has been reported in normal patients undergoing ambulatory urodynamics. The finding does not rule out the

diagnosis of IC/BPS, and treatment of this finding would be unlikely to result in improvement of the patient's bladder pain.

28. **e. All of the above.** IC/BPS should be considered in the differential diagnosis of voiding disorders in men accompanied by irritative symptoms and pelvic pain. A rigorous IC/BPS evaluation can be useful in differentiating IC/BPS from bladder carcinoma in situ, functional or anatomic bladder outlet obstruction, and bacterial prostatitis. Many men with IC/BPS have undergone what has proved to be unnecessary and ill-founded bladder neck surgery.
29. **a. Urethral pain.** Patients with preexisting urethral pain would be poor candidates for supravesical diversion with bladder augmentation as upwards of 30% of patients will need to *self-catheterize* due to poor emptying. The presence of extravesical fat stranding is commonly seen on CT scan, is consistent with the panmural inflammatory process, and should have no bearing upon the indications for surgery. HLs are always identified in a supratrigonal position. Concern for other pathologies, that is, CIS, should be raised if trigonal lesions are identified.
30. **a. Risk factors for BPS.** Emotional, sexual, and physical abuse was shown to be a risk factor in the Boston Area Community Health Survey, and this has been borne out in other studies. A Michigan study compared a control group of 464 women with 215 IC/BPS patients and found that 22% of the control group had experienced abuse versus 37% of the patient group. Those with a history of sexual abuse may present with more pain and fewer voiding symptoms. How reliable these data are is not clear, and it would be wrong to jump to any conclusions about abuse in an individual patient. However, practitioners need to have sensitivity for the possibility of an abusive relationship history in all pain patients, and IC/BPS patients in particular. When patients are found to have multiple diagnoses, the rate of previous abuse also increases, and these patients may need referral for further counseling.
31. **e. HL.** Patients with HLs form a distinct subset of those with BPS. They have identifiable endoscopic and pathologic findings, are less likely to have comorbid conditions, tend to be older, and respond

clinically to bladder fulguration and local steroid injection into the lesions.

32. **a. The patient with provoked vestibulodynia and high-tone pelvic floor dysfunction.** In the "bladder-centric" patient, intravesical therapy may improve bladder symptoms, but if other pain generators go untreated, the reduction of pelvic pain may go unnoticed.

Chapter Review

1. IC/BPS may be defined clinically as "an unpleasant sensation (pain, pressure, discomfort) perceived to be related to the urinary bladder, associated with lower urinary tract symptoms of more than six weeks duration, in the absence of infection or other identifiable causes. The pain or discomfort usually mounts with bladder filling and is the factor that typically prompts urinary frequency. The diagnosis of IC/BPS is one of exclusion because there is no specific test or marker that is diagnostic.
2. Prevalence studies for IC/BPS show wide variation with a general upsurge over time likely related to a more "liberalized" definition of the condition.
3. Current epidemiology suggests approximately 3 million females and 2 million males in the USA with IC/BPS symptoms. A female to male ratio of 5:1 is cited, based upon clinical practice.
4. Sexual dysfunction in female IC/BPS may have a profound effect upon their physical and mental QoL, underscoring the importance of identifying and ultimately treating this problem when present.
5. Common overlapping functional disorders such as high-tone pelvic floor dysfunction, overactive bladder syndrome, vulvodynia, and IBS may necessitate a multidisciplinary approach to achieve symptom relief.
6. The MAPP research network has demonstrated two distinct pain phenotypes in the IC/BPS patient: those with pelvic pain only and those with "pelvic pain and beyond," the latter group representing

over 75% of patients. This suggests that many patients have pain derived from the central nervous system.

7. Fibromyalgia, irritable bowel syndrome, chronic fatigue syndrome, and atopic allergic reactions are overrepresented in the IC/BPS population.
8. High-tone pelvic floor dysfunction has been identified in 70% to 94% of IC/BPS patients either on the basis of urodynamically proven voiding dysfunction or reproduction of pelvic pain upon examination of the pelvic floor musculature.
9. The Hunner lesion (formerly termed “Hunner ulcer”) is the pathognomonic finding for the “classical form” of IC and represents a minority of the IC/BPS population. HLs are easily identified during office cystoscopy, although bladder distention beyond functional capacity and other imaging strategies have been used to enhance identification. Mounting evidence suggests that BPS is indeed a syndrome and distinct from those patients presenting with HLs.
10. There are no reliable differences in clinical presentation between IC/BPS patients with or without HLs.
11. IC/BPS is chronic, the symptoms wax and wane, and remissions are common. Symptom flares are identified in the vast majority of patients, most commonly lasting for days. Common triggers are dietary factors and physical activities, that is, sexual relations, exercises.
12. Some men diagnosed with chronic prostatitis/chronic pelvic pain syndrome, Category III, may also meet criteria for an IC/BPS diagnosis.
13. Prodromal symptoms related to pelvic pain and irritative voiding are often experienced decades before a formal diagnosis.
14. A childhood presentation for IC/BPS is extremely rare. The average age of onset is 40 years.
15. Adult female first-degree relatives of patients with IC/BPS may have a prevalence of IC/BPS 17 times that found in the general population.
16. Although ketamine abuse may be seen at any age, the clinician should be especially wary of the young patient who presents with a

clinical picture of IC/BPS with or without visible inflammation of the bladder wall.

17. There is no histology pathognomonic of IC/BPS. While inflammatory features can be seen in 24% to 76% of patients, histological features are often bland. HLs show a dense, often panmural lymphoplasmacytic infiltrate. The current role of biopsy in the management of the IC/BPS patient is in the patient with HLs to rule out other pathology such as carcinoma in situ.
18. Antiproliferative factor (APF) is secreted by bladder epithelial cells, inhibits bladder epithelial cell proliferation, and is used as a marker of the disease. It may be the primary cause of syndrome in some patients. Urine APF appears to have the highest sensitivity and specificity of the markers studied for this disease.
19. Current studies suggest that the etiology of IC/BPS symptoms may vary. Local processes such as “leaky epithelium,” mast cell activation, and neurogenic inflammation may produce symptoms for some patients; however, the presence of co-morbid widespread pain syndromes, regional pain syndromes, and immune-mediated pain in the majority of patients suggest that more global pathologies may also be responsible.
20. Cross-sensitization among pelvic structures, that is, bowel and bladder, may contribute to chronic pain syndromes because this may result in alteration in function of adjacent pelvic organs.
21. Bladder compliance in patients with IC is normal; however, capacity is often reduced and pain/discomfort is reproduced with filling.
22. Failure to diagnose other causes of pelvic pain, that is, high-tone pelvic floor dysfunction, vulvodynia, that may occur with or without IC/BPS is one of the most common reasons for a suboptimal therapeutic outcome.
23. Microscopic hematuria and/or pyuria are not reliable findings to predict the presence of an HL. Office cystoscopy should be considered if an HL is suspected or the patient is not responding to empiric therapy.
24. Although hydrodistention of the urinary bladder is not part of the standard diagnostic evaluation, recent findings suggest that a low bladder capacity at the time of hydrodistention (anesthetic bladder

capacity) may represent a distinct clinical phenotype, having higher symptom scores and fewer reports of depression and irritable bowel syndrome, essentially a more bladder-centric patient picture. The presence of glomerulations (puntate submucosal hemorrhages identified during hydrodistention) is not necessary for diagnosis. The finding lacks sensitivity and specificity.

25. Conservative treatment strategies are recommended with progression to more aggressive care in the setting of poor symptom control and an unacceptable quality of life. Initial aggressive care may be indicated for patients who present with severe symptomatology that, in the clinician's opinion, is unlikely to have an adequate response to more conservative measures.
26. Patient education, dietary manipulation, nonprescription analgesics, and pelvic floor relaxation sensation techniques constitute the initial treatment of BPS.
27. Catastrophic thinking, the irrational, consuming fear of a disastrous outcome, appears to enhance the perception of pain, but may be modified with cognitive behavioral therapy.
28. Many patients find their symptoms adversely affected by certain food/beverage groups. A step-wise method to determine dietary sensitivities such as an elimination diet may play an important role in patient management.
29. Pentosan polysulfate sodium (an oral agent) and dimethylsulfoxide (an intravesical agent) are the only two FDA-approved medications for the treatment of IC/BPS symptoms.
30. Amitriptyline, a tricyclic antidepressant, is not FDA approved for the treatment of IC/BPS. Nevertheless, it remains a staple of oral treatment given its antihistaminic properties, its ability to promote sleep, and reduce neuropathic (central and peripheral) pain.
31. Intravesical agents may improve symptoms of IC/BPS but would have limited success in those patients with centralized pain. The frequency and duration of therapy, and the value of maintenance therapy have not been well studied.
32. Sacral neuromodulation and intradetrusor injection of botulinum toxin A are currently considered as fourth-line treatment strategies. Although urinary retention is a relatively rare event with BTX-A,

caution should be exercised when considering its use in the patient with obstructive voiding symptoms.

33. Major reconstructive procedures may be considered when all conservative forms of management have failed. Diversion, and even cystectomy with diversion, cannot guarantee a pain-free result, and it is critical for the patient to factor this into their decision. Patients with a clear-cut source of bladder pain (such as HLs) with a small-capacity, poorly compliant bladder are most likely to have a favorable clinical outcome.
34. Substitution cystoplasty and continent diversion both fail in some IC/BPS patients because of the development of pain in the bowel segment used or contraction of the bowel segment.